

Andrew Kim

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Interdisciplinary software engineer looking for opportunities in back-end, front-end, and/or full-stack, with additional experience in simulation / visualization software and game development

Work Experience

Frontend Developer

Seattle, WA

Freelance — Website Development

January 2026 – Present

- Partnered with small businesses to design and build responsive, modern websites using TypeScript and React, translating client needs into clean, maintainable codebases.
- Migrated client sites from platforms like GoDaddy and IONOS, auditing and eliminating unnecessary services to reduce overhead costs by up to but not limited to \$1,200/month.
- Established modular custom component libraries to minimize technical debt and improve long-term maintainability for non-technical owners.
- Improved search visibility through a combination of technical SEO practices and targeted keyword research.
- Ensured inclusive, accessible design across all projects by adhering to ADA compliance standards and responsive design principles.

MoodRoom

Seattle, WA

Lead Game Developer — Game Development

May 2025 – December 2025

- Conceptualized and built a minimum viable product (MVP) that effectively demonstrated product potential and secured interest from venture capital investors, raising over \$100,000 in funding.
- Responsible for building and porting mobile app into multiple platforms, including iOS and Android, creating 1000+ wishlists and 4,000+ beta sign-ups during early access.
- Directed technical decision-making across a 5-person cross-functional team (design, art, and engineering), balancing innovation with scope management to ensure 95% of sprint deadlines were met on time.
- Architected and developed core gameplay systems from the ground up using the Godot engine, delivering performant and engaging player experiences.

Schindler Iterative Translation Lab, VA Puget Sound Medical Center

Seattle, WA

Data Engineer — Machine Learning and Data Visualization

January 2025 – June 2025

- Processed and cleaned 1.2+ TB of mouse behavioral data to visualize polysubstance use in mice, enhancing data interpretation and downstream efficiency by 35%.
- Selected and implemented a variety of machine learning models, including K-Means clustering, Multiple Correspondence Analysis (MCA) and Principal Component Analysis (PCA), reducing feature dimensionality by ~70% while preserving behavioral variance.
- Designed and deployed interactive data visualizations including dendrograms, heatmaps, and 15+ high-impact dashboards to facilitate improved decision-making for the research team.
- Co-authored a peer-reviewed research paper by contributing critical data engineering and visualization methodologies.

Lizard Tank, LLC.

Seattle, WA

Lead Game Developer — Game Development

July 2024 – Present

- Architected, designed, and implemented a custom physics system, saving the company thousands of dollars by eliminating the need for third-party solutions, while ensuring realistic and dynamic object interactions for enhanced gameplay.
- Developed and optimized user interface (UI) systems, reducing user frustration and saving hours of potential disengagement, resulting in increased player satisfaction and significantly boosting overall game and product engagement.
- Engineered AI behaviors using state machines and advanced algorithms for path planning and obstacle avoidance, leading to intelligent and responsive non-player character (NPC) movement.

MathWorks Inc.

Natick, MA

Software Engineer — Engineering Development Group

May 2022 – November 2023

- Architected and built a back-end research tool that enabled customers to efficiently sift through thousands of third-party analog devices and automatically generate relevant specification plots, saving over 1000 man hours of manual web searching.
- Built the first proof of concept, integrating Simulink with Unreal Engine for a leading construction firm, enabling virtual testing that saved ~\$1M per vehicle on real-world testing and potential equipment damage.
- Led as Scrum Master and Operations Lead a team of 50+ engineers to efficiently assist hundreds of industry leading companies and academic institutions in defense, tech, biomedical, etc., with troubleshooting MATLAB and Simulink models as quickly and accurately as possible.

Technical Skills

Programming — C++, MATLAB, Python, Java, C#, GDScript, React

Software — Simulink, LabView, Microsoft Office, Perforce, GitHub, Git, Visual Studio, Godot, Unity

Mechanical — SolidWorks, Catia v5, Autodesk Inventor

Publications & Certifications

- Anxiety and risk-taking behavior maps onto opioid and alcohol polysubstance consumption patterns in male and female mice
<https://www.biorxiv.org/content/10.1101/2024.08.22.609245v2.full>

- AWS Machine Learning Specialist

Education

Western Governors University

Master of Science in Computer Science and Artificial Intelligence (MSCS)

Remote / Salt Lake City, UT

May 2025 – December 2025

Purdue University

Bachelor of Science in Astronautical and Aeronautical Engineering (BSAAE)

West Lafayette, IN

May 2022